Introduction to Demographic Methods

(4: Population projections)

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Introduction to Demographic Methods

Universität Rostock

1 – Introduction. Why study Demography? . . . Git, version control, course materials GitHub
2 – Mortality
3 – Fertility
4 – Population projections
5 – Migration
~~6 – Kinship~~
7 – The future, and Digital and Computational Demography
9 – Wrap up + Description of examination
10 – Final examination: 3 min presentation to share your selected topic + 2 min Q&A; 6 weeks to prepare a short essay and empirical analysis

Discussing previous week's materials

1) Slides, Readings

1. Any questions on my slides?
2. Let's discuss the "key reading material".
3. Example: Discuss at least 1-2 points, e.g., 1) What was surprising and counterintuitive for you? 2) What did you find that was already expected? 3) What could be extended in this reading/work?
4. Any thoughts about the additional reading materials?

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2) Hands-on

1. Any problems in installing the requirements?
2. Any issues in running the codes in hands-on materials to reproduce the examples?
3. Did anyone extend the examples? How? What did you do and find?

Few points

- Difference between stable and stationery populations, MAlexander slides, P6
- The idea and goals of population project, MAlexander slides, P23
- Amaral slides, P10, "projecting the size and age distribution of a population forward through time are tables called "Leslie matrices""
- "What is being projected with Leslie matrices are expected numbers of individuals rather than probabilities"
- Important note on Leslie matrice that columns are starting age group (start from), and rows are the age group where one can survive into (end up into).
- As always, check the chapters in Wachter (2014) for more details and examples.

Switch to Ernesto Amaral's slides: "Lecture05" and Monica Alexander's slides: "5_pop_proj"

Hands-on part

Needed installations (or checking requirements)

For today's hands-on part, see the Readme file under "03_MonicaALabs" folder and download the requested files under "data" folder.

We will go over the notebook called "5_pop_proj_edits.qmd" and see what it does. You need to activate Renv and install R packages listed in the first code block of the notebook, i.e., "tidyverse, here, readxl, janitor" (please check the notebook for the latest list).

Needed: R, RStudio, and Renv environment.

Where to learn basics of R and Python?

To start with Python

Check this website first:
https://www.w3schools.com/python/python_intro.asp

Check this repository by Vincent Traag and others, for an introductory course and code:
https://github.com/vtraag/intro-python

Or this one by Data Carpentry:
https://datacarpentry.org/python-ecology-lesson/

To start with R

Check this website first:
https://www.w3schools.com/r/default.asp

This "very short introduction to R" is a good start:
https://cran.r-project.org/doc/contrib/Torfs+Brauer-Short-R-Intro.pdf

Or this course by Data Carpentry:
https://datacarpentry.org/R-genomics/index.html

Reading materials for today

Key reading material, Chapters 5 and 10

Wachter, K. W. (2014, June). Essential Demographic Methods. Harvard University Press.

Additional reading materials

Alexander, M. (2026). Advanced Data Analaysis course in Demographic Methods. Retrieved April 8, 2026, from https://github.com/MJAlexander/soc6708

Preston, S., Heuveline, P., & Guillot, M. (2000). Demography: Measuring and Modeling Population Processes. Wiley.

What to do before the next session?

1) Slides, Readings

1. Study my slides carefully.
2. Check the slide with reading materials.
3. Read the "key reading material".
4. Prepare at least 1-2 discussion points, e.g., What was surprising and counterintuitive for you? What did you find that was already expected? What could be extended in this reading/work?
5. Skim over the additional materials.

2) Hands-on

1. Check the slide and folder for hands-on materials.
2. Install the requirements as outlined.
3. Run the codes in hands-on materials to reproduce the examples, and if you can extend them.
4. Note down the unclear points, errors, and questions for the next session.

Thanks for your attention!

Questions and comments are always welcome!

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LinkedIn/BlueSky/Twitter/Mastodon: @akbaritabar

References

Wachter, K. W. (2014, June). Essential Demographic Methods. Harvard University Press.